



5G-SNPA System

Smart Number Plate Authorization System for Security Purposes

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Abstract

This project tries to automate **vehicle authorization** and logging by reading number plates from vehicles in real-time using a **5G-camera**.

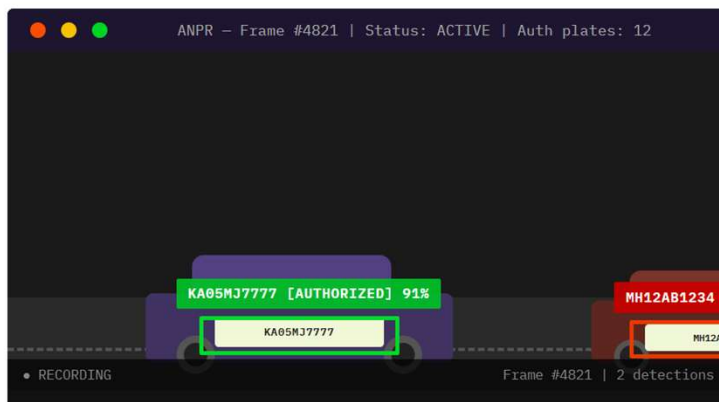
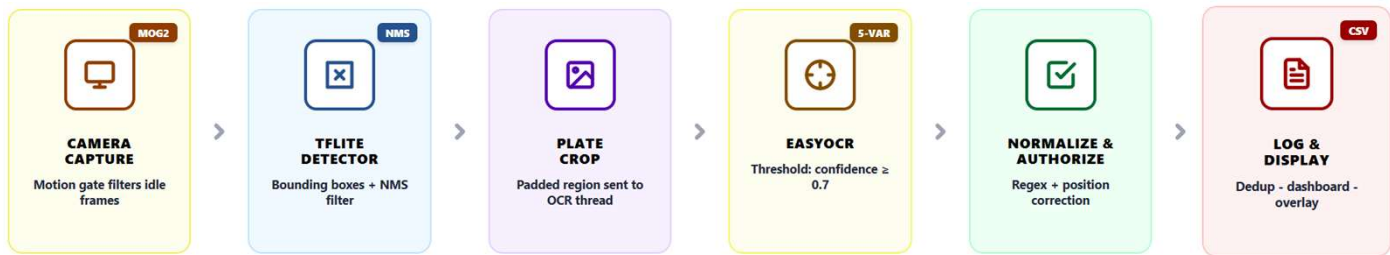
Main Results

Our system performs the complete access-control pipeline:

feed → detect → read → authorize → log

in real time on a standard laptop, with a web-based admin dashboard accessible from any browser on the same network.

System Pipeline



Live Camera Output:

Shows real-time video with detected plates, status, and confidence overlaid on the frame.

POSITION-AWARE OCR CORRECTION

K A 0 5 M J 7 7 7 7

STATE DISTRICT SERIES SERIAL

ALPHA pos: digit→letter
(0→0, 1→I)

DIGIT pos: letter→digit
(0→0, 8→8)

SNPA Admin Dashboard

Plates file: c:\users\93880\Desktop\anpr_improved\authorized_plates.txt

Auto Refresh

Dashboard

Output:

Displays recent detections and logs, and allows managing authorized plates with instant updates.

Authorized Plates

Add new plate

MH12AB1234

Add

4 authorized

Manage list

DL01XY9999 Remove

KA05MJ7777 Remove

Access Logs

timestamp	plate	status	ocr_confidence	frame_no
2026-04-28 11:52:35	LL92P8106	UNAUTHORIZED	0.600100	160
2026-04-28 11:50:02	RJ1LCV0002	UNAUTHORIZED	0.421900	4035
2026-04-28 11:49:45	DL26CA1111	UNAUTHORIZED	0.999600	3640
2026-04-28 11:49:29	DL1AB2345	UNAUTHORIZED	0.949200	3245
2026-04-28 11:49:01	DL09U9111	UNAUTHORIZED	0.679200	2515
2026-04-28 11:48:32	AR26DK8337	UNAUTHORIZED	0.525200	1805
2026-04-28 11:48:30	HR26DK8337	UNAUTHORIZED	0.499800	1780
2026-04-28 11:47:31	AS8YQ7167	UNAUTHORIZED	0.003900	725
2026-04-28 11:42:16	MH20EE7598	UNAUTHORIZED	0.888800	880
2026-04-28 11:41:56	RJ1LCV0002	UNAUTHORIZED	0.558700	560